SHUAIB IQBAL

AI Engineer | Machine Learning, LLM, RAG & Agentic AI Systems

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PROFESSIONAL SUMMARY

AI Engineer with 4+ years building production AI systems — LLM-powered applications, RAG pipelines, agentic workflows, and the ML models and scalable backends that power them — for enterprise clients across FMCG, IoT, and SaaS. Experienced across the full stack: self-hosting and fine-tuning open-source models (LLaMA 3 via Ollama), designing FastAPI/Django REST APIs, and building Docker-containerized data pipelines. Delivered Revenue Growth Management AI for global brands including EBM, PMI, RB-PK, and Unilever, cutting out-of-stock events by 20% and driving double-digit commercial growth.

TECHNICAL SKILLS

LLM & Generative AI: OpenAI API · LangChain (LCEL, Chains, Runnables, Memory) · HuggingFace Transformers · Ollama (self-hosted/fine-tuned models) · LLaMA 3 · Prompt Engineering & Structured Output

Agentic AI & RAG: LangGraph · CrewAI · Multi-Agent Systems · Tool-Calling Agents · FAISS · Pinecone · ChromaDB · Semantic Search

Machine Learning & Deep Learning: Scikit-learn · TensorFlow · PyTorch · XGBoost · LSTM · CNN · Time-Series Forecasting (ARIMA, Prophet) · Anomaly Detection

Backend & Data Engineering: FastAPI · Django REST Framework · Flask · PostgreSQL · SQLAlchemy · REST API & Microservices Design · ETL Pipelines

Cloud & DevOps: AWS (EC2 — compute & deployment) · Docker · Git · Linux · Streamlit · Plotly · Jupyter · Cursor AI · Claude AI

WORK EXPERIENCE

AI/ML Engineer *Oct 2022 – Present*

- Built a natural-language-to-SQL tool (OpenAI API + LangChain) enabling plain-English querying over a 14M-row FMCG sales dataset, eliminating manual reporting cycles.
- Developed an AI-first Sales Intelligence Chat platform (OpenAI API + LangChain Agents + Streamlit) with multi-turn session memory, live sortable tables, and follow-up chaining for real-time FMCG insights.
- Delivered Revenue Growth Management AI pipelines for EBM, PMI, RB-PK, and Unilever — predictive selling models, route optimization, and White Space Identification algorithms driving double-digit commercial growth, and reduced out-of-stock events by 20% through time-windowed stockout forecasting and automated alerting.
- Built multi-system integrations — REST APIs (FastAPI/Django) exposing ML insights to SaaS clients — and ETL pipelines (Python, SQL) unifying POS and census data across millions of records, while owning delivery timelines across concurrent, Docker-containerized projects.

Associate Machine Learning Engineer *May 2022 – Oct 2022*

- Improved HVAC forecasting accuracy by 25% by building and benchmarking ARIMA, LSTM, and XGBoost models on multi-sensor time-series data, selecting the best performer per building zone.
- Built real-time anomaly detection systems for IoT energy-wastage reduction, flagging abnormal consumption patterns and cutting manual monitoring effort for the facilities team.
- Designed time-series databases and backend services to ingest, clean, and store high-frequency sensor data from multiple IoT devices, ensuring reliable data for downstream forecasting models.
- Built Streamlit/Plotly dashboards giving non-technical stakeholders live visibility into forecasts and anomalies, and automated REST API reporting pipelines to remove manual report generation, while collaborating with hardware and backend engineers to validate sensor data quality.

ML Engineer — Internship *Feb 2022 – Apr 2022*

- Built sales forecasting models (Prophet, LSTM, XGBoost) on historical transaction data, evaluated with MAE, RMSE, and R^2 to select the most reliable model per product category.
- Performed RFM (Recency, Frequency, Monetary) customer segmentation to identify high-value customer groups and support targeted sales strategies.
- Built interactive dashboards and backend reporting scripts to present forecasts and segmentation results to the team.

PROJECTS

Sales Intelligence Chat — Ask Your Data Anything — *LangChain · OpenAI API · GPT-4 · Streamlit*

- LangChain agent querying a 14M-row FMCG dataset in plain English, with multi-turn session memory, follow-up chaining, and live sortable table rendering.
- Designed the agent's tool-calling logic to translate natural-language questions into safe, validated SQL queries before execution.

Portfolio AI Assistant — RAG System — *LangChain · FAISS · OpenAI API · FastAPI · Render*

- End-to-end RAG pipeline (document loaders, text splitters, FAISS vector store, OpenAI embeddings) for context-grounded Q&A; deployed live on Render.
- Tuned chunk size and retrieval parameters to balance answer accuracy against response latency for a smooth chat experience.

E-Commerce AI Chatbot (Fully Offline) — *LangChain · LLaMA 3 · Ollama · Flask*

- Fully offline AI chatbot for e-commerce using a self-hosted LLaMA 3 model via Ollama, with RAG-style contextual Q&A and zero cloud dependency.
- Built a lightweight Flask interface so the chatbot could run entirely on local hardware with no external API costs.

Conversational AI Chatbot with Memory — *LangChain · ConversationBufferMemory · Streamlit*

- Stateful multi-turn chatbot with windowed memory, contextual recall, and LCEL chain composition.
- Implemented memory-window tuning to keep conversation context relevant while controlling token usage on longer sessions.

AI-Powered Document Q&A Bot — *LangChain · FAISS · OpenAI API · Streamlit*

- Full RAG pipeline to upload any PDF and ask questions in plain English — document loading, chunking, FAISS vector indexing, and retrieval-augmented generation.
- Added source-highlighting so answers are traceable back to the exact passage in the uploaded document.

AI-Powered Resume Assistant (Fully Offline) — *LangChain · LLaMA 3 · Ollama · Streamlit*

- Privacy-first offline LLM assistant for resume generation and contextual rewriting, using self-hosted local models, prompt chaining, and structured output design.
- Designed prompt templates that keep generated resume content consistent in tone and formatting across sections.

EDUCATION

MS Computer Engineering *2018 – 2021*
NUST — National University of Sciences & Technology, Islamabad · Thesis: Predicting Residential Energy Consumption via Encoder–Decoder CNN-LSTM

B.Sc. Computer Systems Engineering *2014 – 2018*
IUB — The Islamia University of Bahawalpur · Thesis: Pattern Formation of Swarm Robots

CERTIFICATIONS

- Claude 101 — Anthropic
- AI Fluency Framework & Foundations — Anthropic
- Fundamentals of LLMs of The LLM Course — Hugging Face
- Introduction to ChatGPT — DataCamp
- Data Science with Python — SkillUp Online
- Data Visualization using Plotly — Coursera
- Pandas — Kaggle
- Docker for the Absolute Beginner — KodeKloud
- Python for Data Science — Dataquest.io
- Python Basics & Programming for Everybody — Coursera / University of Michigan